[14]:
```python
df = pd.read_csv('train.csv')
df
```

[14]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows × 12 columns

[22]:
```
df.describe()
```

[22]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

[24]:
```
df.duplicated().sum()
```

[24]:
```
np.int64(0)
```

[17]:
```
df.head()
```

[17]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

]:

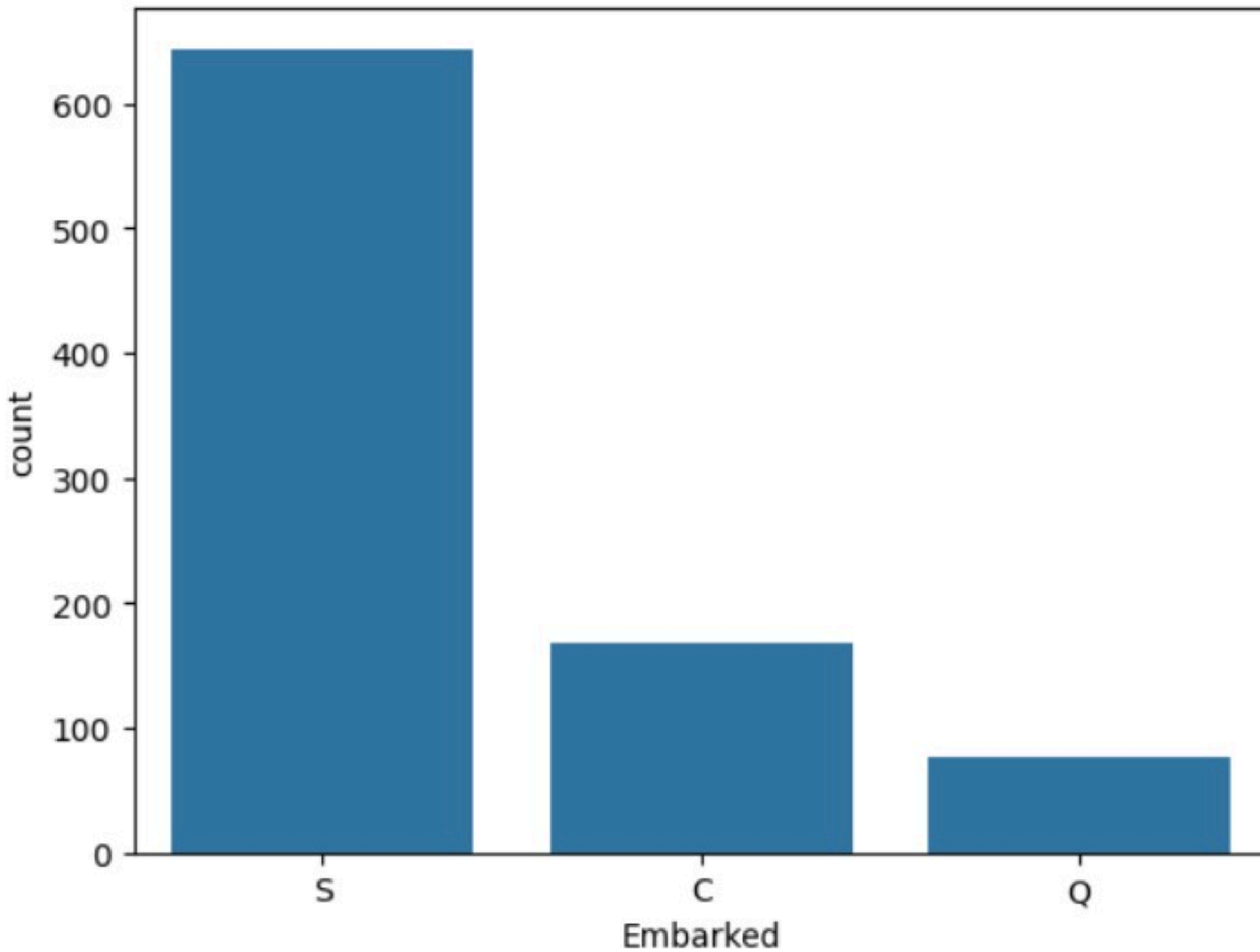
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

[21]:
```
df.isnull().sum()
```

[21]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

[38]:
```
sns.countplot(x = 'Embarked', data = df)
plt.show()
```



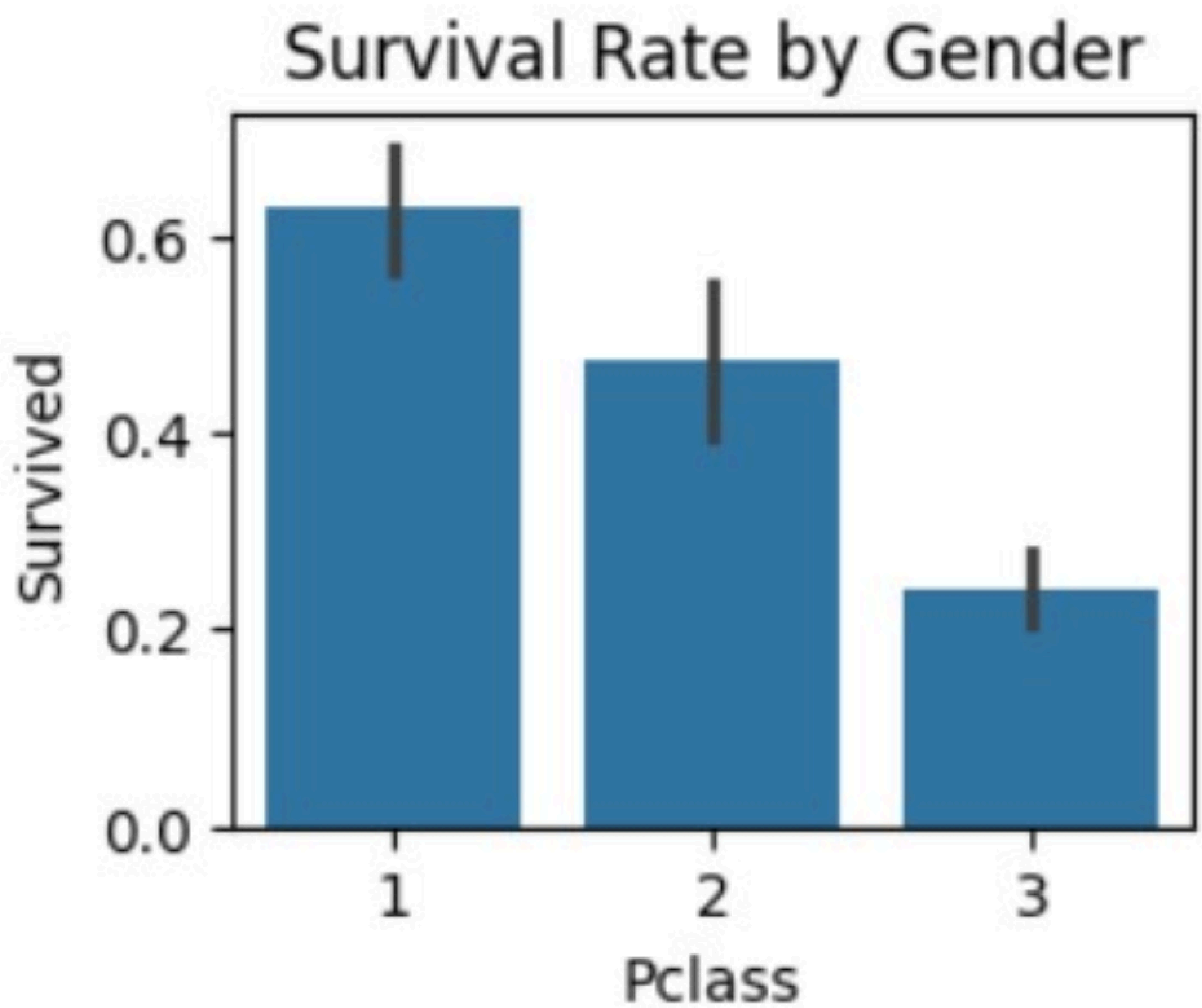
[39]:
```
plt.figure(figsize=(15,10))
```

[39]:
```
<Figure size 1500x1000 with 0 Axes>
```

```
<Figure size 1500x1000 with 0 Axes>
```

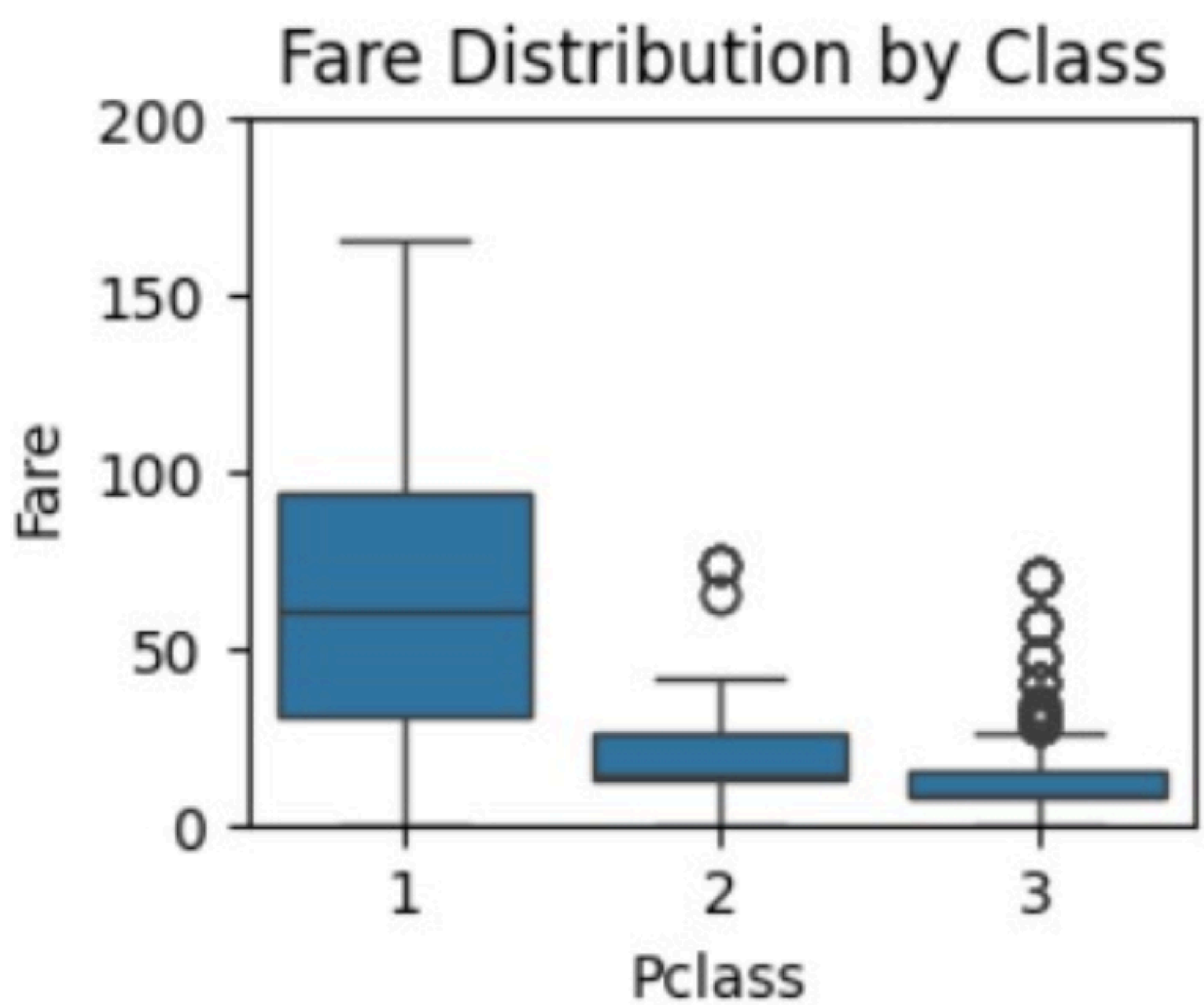
[42]:
```
plt.subplot(2,2,2)
sns.barplot(x='Pclass',y='Survived',data=df)
plt.title('Survival Rate by Gender')
```

[42]:
```
Text(0.5, 1.0, 'Survival Rate by Gender')
```



[45]:
```
plt.subplot(2,2,4)
sns.boxplot(x='Pclass',y='Fare',data=df)
plt.ylim(0,200)
plt.title('Fare Distribution by Class')
```

[45]:
```
Text(0.5, 1.0, 'Fare Distribution by Class')
```

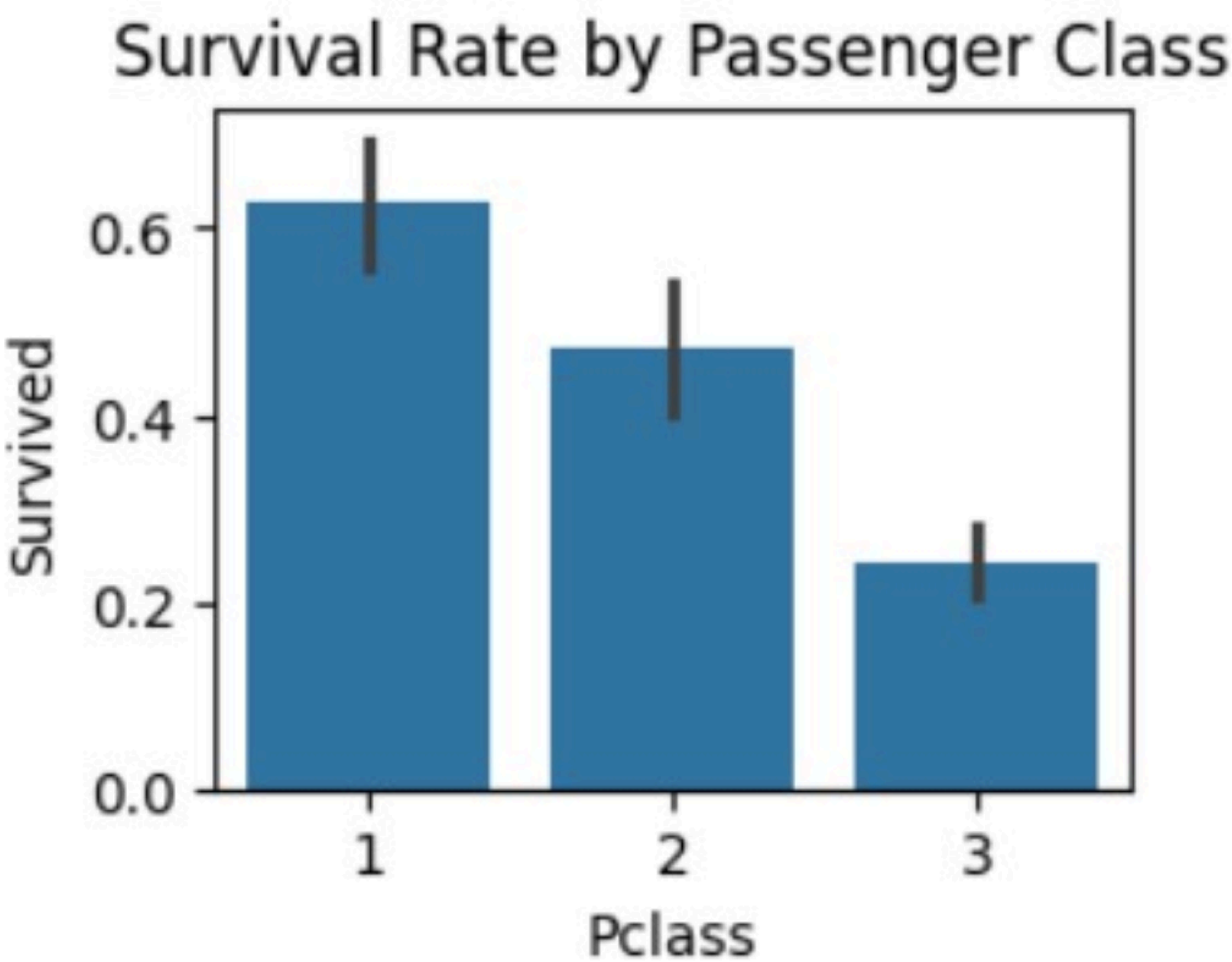


[46]:
```
plt.tight_layout()
plt.show()
```

```
<Figure size 640x480 with 0 Axes>
```

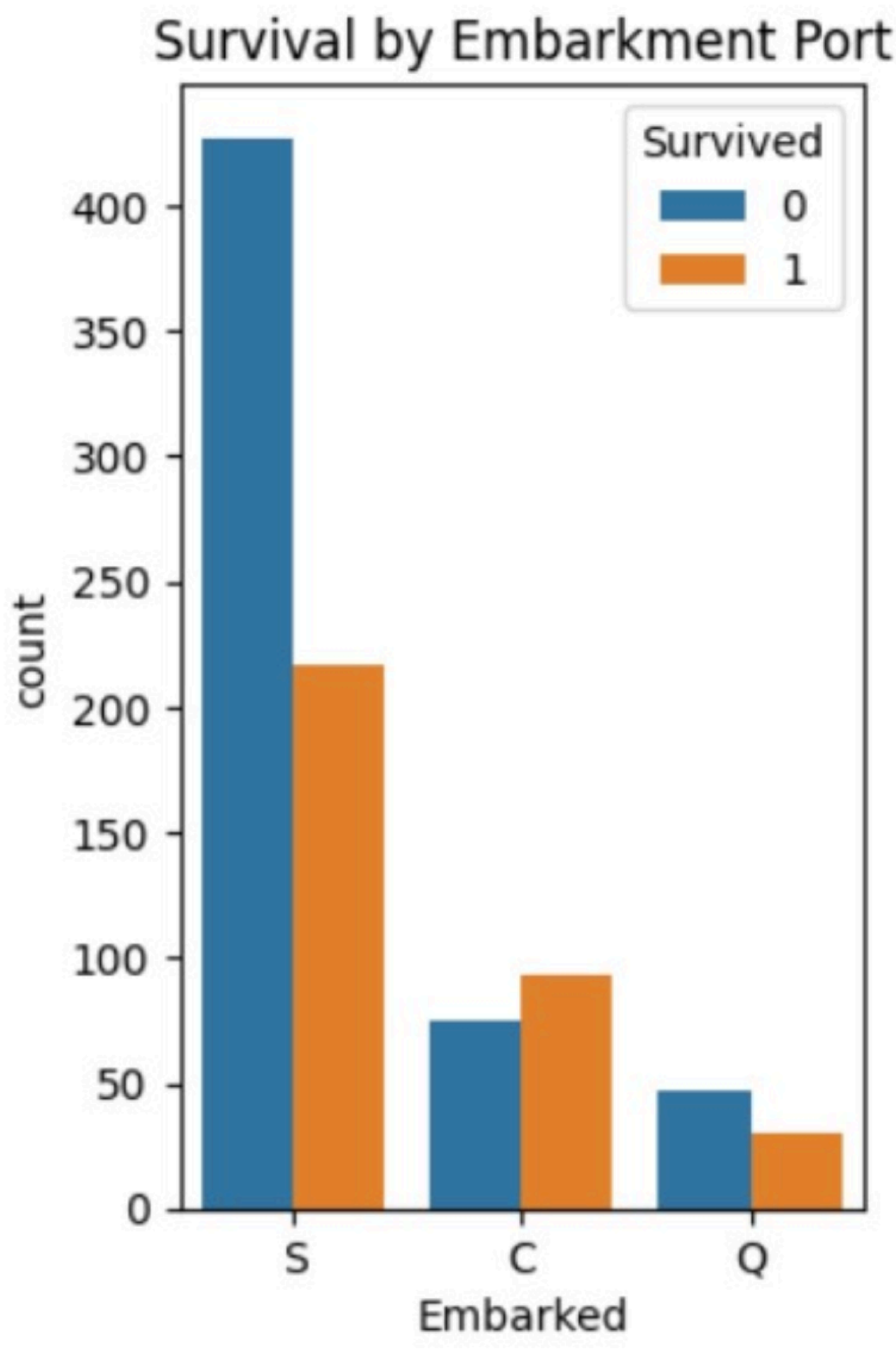
[43]:
```
plt.subplot(2,2,1)
sns.barplot(x='Pclass',y='Survived',data=df)
plt.title('Survival Rate by Passenger Class')
```

[43]: Text(0.5, 1.0, 'Survival Rate by Passenger Class')



[52]:
```
plt.subplot(1,2,2)
sns.countplot(x='Embarked',hue='Survived',data=df)
plt.title('Survival by Embarkment Port')
```

[52]: Text(0.5, 1.0, 'Survival by Embarkment Port')



[53]:
```
plt.tight_layout()
plt.show()
```

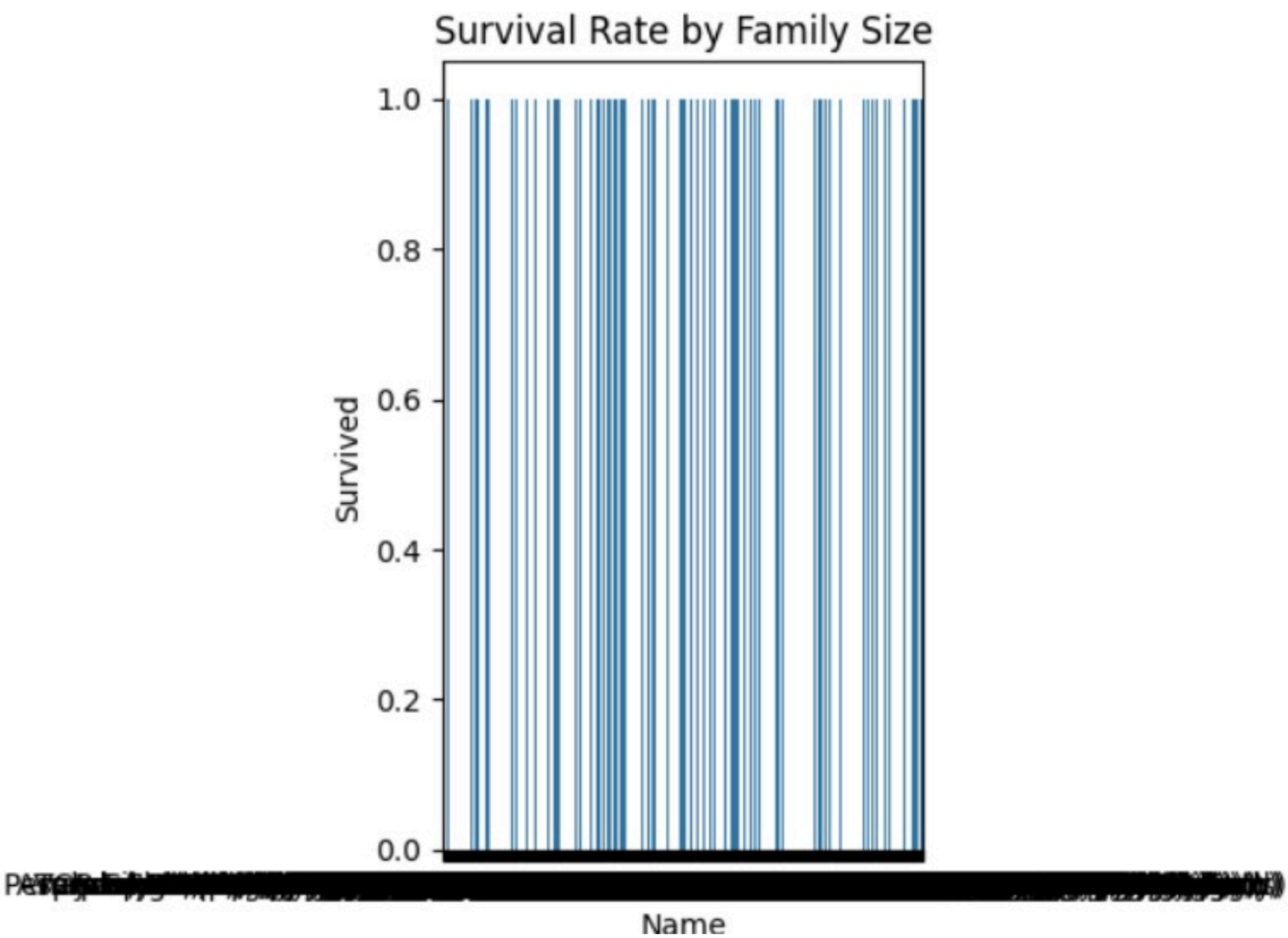
<Figure size 640x480 with 0 Axes>

[50]:

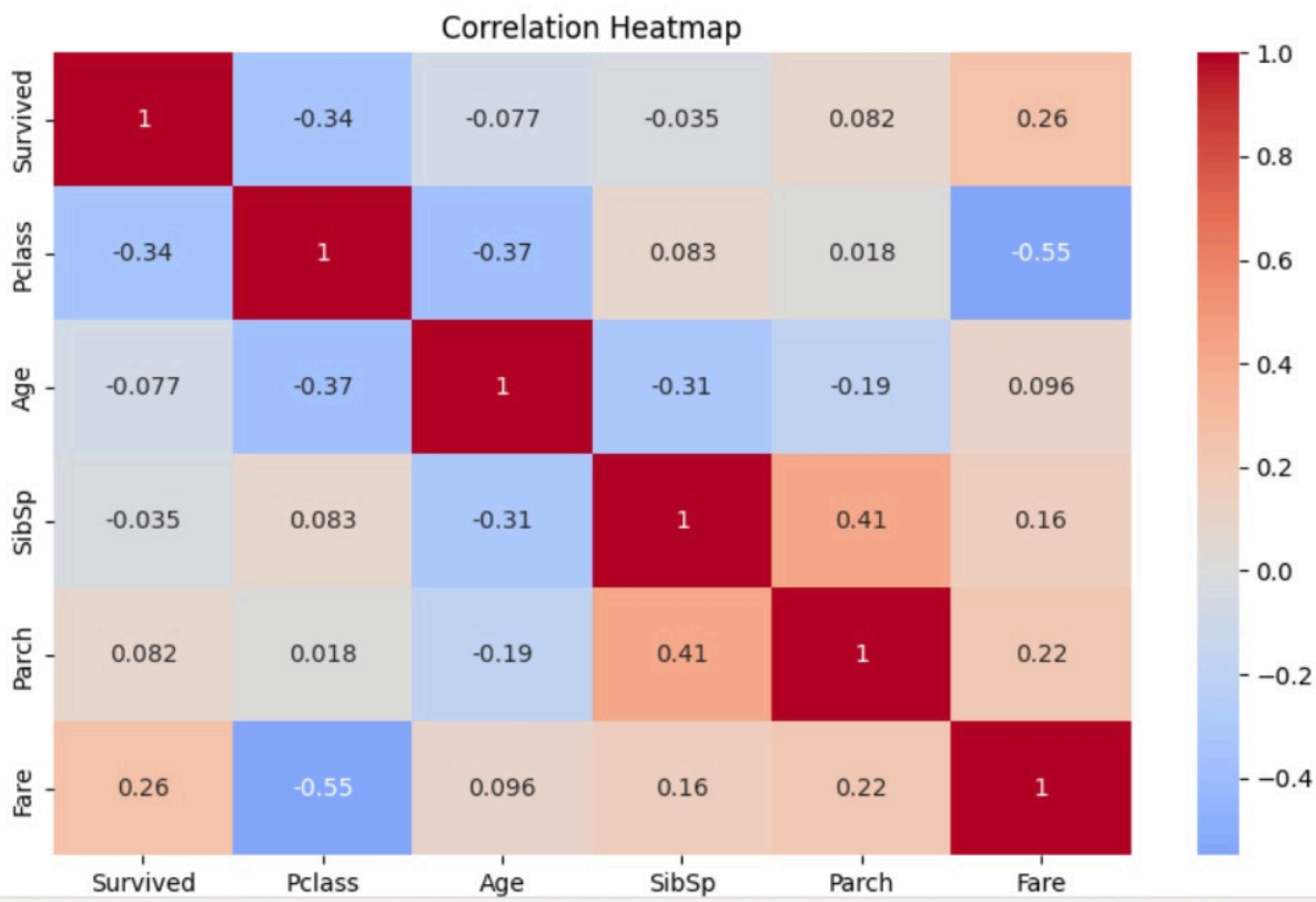
```
plt.subplot(1,2,1)
sns.barplot(x='Name',y='Survived',data=df)
plt.title('Survival Rate by Family Size')
```

[50]:

```
Text(0.5, 1.0, 'Survival Rate by Family Size')
```



[54]:
```
plt.figure(figsize=(10, 6))
numeric_cols = ['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']
sns.heatmap(df[numeric_cols].corr(), annot=True, cmap='coolwarm', center=0)
plt.title('Correlation Heatmap')
plt.show()
```



[55]:

```python
print("\n===== Key Findings =====")
print(f"1. Overall survival rate: {df['Survived'].mean():.2%}")
print(f"2. Female survival rate: {df[df['Sex']=='female']['Survived'].mean():.2%}")
print(f"3. Male survival rate: {df[df['Sex']=='male']['Survived'].mean():.2%}")
print(f"4. 1st class survival rate: {df[df['Pclass']==1]['Survived'].mean():.2%}")
print(f"5. 3rd class survival rate: {df[df['Pclass']==3]['Survived'].mean():.2%}")
print("6. Children (<10) had higher survival rates than adults")
print("7. Higher fare passengers had better survival chances")
print("8. Passengers with 1-3 family members had highest survival rates")
```

```
===== Key Findings =====
1. Overall survival rate: 38.38%
2. Female survival rate: 74.20%
3. Male survival rate: 18.89%
4. 1st class survival rate: 62.96%
5. 3rd class survival rate: 24.24%
6. Children (<10) had higher survival rates than adults
7. Higher fare passengers had better survival chances
8. Passengers with 1-3 family members had highest survival rates
```

[ ]: